

pylhe: A Lightweight Python interface to Les Houches Event files

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Summary

pylhe is a lightweight Python library that provides a simple and efficient interface for reading and writing Les Houches Event (LHE) files, a standard format used by Monte Carlo event generators in high-energy physics (Alwall & others, 2007). The library enables memory-efficient streaming of events from XML format (.lhe, compressed .lhe.gz) and HDF5 format (.h5, .hdf5, lheh5) files through a pythonic iterator interface, allowing researchers to process arbitrarily large event files without loading all events into memory simultaneously. It is a standalone, event-generator-independent package, installable from PyPI and Conda. pylhe, part of the Scikit-HEP community project (Rodrigues & others, 2020; The Scikit-HEP Project, 2016), is well maintained and integrated within the broader scientific Python ecosystem.

Historically, the first standards for event representation in high-energy physics were the HEPEVT and HEPRUP common blocks (Boos & others, 2001), which provided a Fortran-based structure for storing event information. As the complexity of Monte Carlo event generators increased, the need for a more flexible and extensible format led to the development of the LHE file format. LHE introduced an XML structure that allowed for better organization of event data and facilitated interoperability between different tools. Typically, LHE files are used to describe parton-level events generated by matrix element generators that are passed to parton shower and hadronization programs producing HepMC files (Buckley et al., 2021; Dembinski et al., 2022; Dobbs & Hansen, 2001), or directly to analysis frameworks such as Rivet (Bierlich et al., 2024; Bierlich & others, 2020; Buckley et al., 2013). In contrast to LHE files, HepMC stores fewer details about the particles, but also contains many more particles per event. Both files are plain text-based formats, making them human-readable but potentially large. Consequently, compression using the gzip library is common practice for both formats.

Since pylhe version 2.0.0, the library also supports the HDF5-based (The HDF Group, 1998) LHEH5 format (Bothmann et al., 2024; H"ocher et al., 2019), which allows for significant improvements in write performance while keeping file sizes for event storage under control.

The LHE format stores information within <init> and multiple <event> blocks consisting of whitespace-separated values designed historically for straightforward parsing in Fortran. Further details can be found in the original definition of the Les Houches Event file standard (Alwall & others, 2007). Following the original publication there were two extensions to the LHE format, version 2.0 in 2009 (Butterworth & others, 2010) and version 3.0 in 2013 (Andersen & others, 2014). pylhe completely implements the widely adopted version 3.0, that is the addition of <scales>, <generator>, and multiple weights via <initrwgt>, <rwgt>, <weight>, <weights>, <wgt>, and <weightgroup> on top of the version 1.0. Features that were introduced in version 2.0 but removed in version 3.0, such as <pdfinfo> or <clustering>, are not supported.

Statement of need

The LHE format is used by all major Monte Carlo event generators such as MadGraph (Alwall et al., 2014), POWHEG-BOX (Alioli et al., 2010; Frixione et al., 2007; Nason, 2004), Sherpa (Bothmann & others, 2019; Gleisberg et al., 2009), HERWIG (Bahr & others, 2008; Bellm & others, 2016, 2020; Bewick & others, 2024; Corcella et al., 2001), Pythia (Bierlich & others, 2022; Sjostrand et al., 2006, 2008; Sjöstrand et al., 2015), Whizard (Kilian et al., 2011; Moretti et al., 2001). While interfaces for C/C++/Fortran exist in the respective generators, a lightweight, standalone and easy-to-use Python interface equally supporting all these event generators was missing until the inception of pylhe in 2015. Additionally, pylhe can serve as a crucial interface for emerging machine learning applications in particle physics, allowing researchers to efficiently extract event data for training neural networks and other machine learning models used in event classification, anomaly detection, and physics analysis. Past experience and the ever-growing number of publications citing pylhe have shown that the portfolio of users goes from experimentalists to phenomenologists, some of which are themselves developers of event generators.

As a library, pylhe is straightforwardly installable via the standard means, and is maintained within the Scikit-HEP community project that is very well integrated within the broader scientific Python ecosystem.

State of the field

Unlike the existing C/C++/Fortran interfaces provided by the Monte Carlo event generators, pylhe offers a standalone, pure Python interface that is more accessible and easier to use. When pylhe was first developed, there were no other standalone Python libraries available for reading and writing LHE files. However, a Python parser is distributed as part of MadGraph (Alwall et al., 2014), while MadAnalysis 5 exposes its C++ LHE reader and writer through a Python interface (Conte et al., 2013). Nowadays, there are a few other smaller Python libraries with less adoption than pylhe, which provide only read functionality and are no longer actively maintained, such as lhereader (Albert, 2021; Biswas, 2017). For completeness, it should be mentioned that several LHE libraries exist in other programming languages, such as Go (go-hep, Binet et al., 2017), Rust (lhe, Fukuda, 2018; lhef, Maier, 2024b; Weber, 2018; event_file_reader, Maier, 2024a), Julia (LHEF.jl, Fischer & Ling, 2023) and Haskell (lhe.hs, Lawrence, 2019). These provide varying degrees of completeness, but none support the LHEH5 format yet.

Software design

pylhe allows for easy reading and writing of .lhe and .lhe.gz files in Python, enabling seamless integration into modern data analysis workflows in high-energy physics. The pythonic event yielding approach allows for memory-efficient processing of arbitrarily large LHE files by streaming events one at a time rather than loading all of them at once into memory. Internally, pylhe uses xml.etree.ElementTree to parse the XML structure, since using the lxml library did not provide a significant speed up. For LHEH5 files the library uses h5py (Collette, 2013) to read and write the HDF5 format, via the same pythonic data structures, making it easy to convert between the standard XML and newer HDF5 formats.

The library facilitates quick validation of event files through programmatic access to event structure and particle properties, making it straightforward to perform sanity checks on generated events. This can be done for example via the integration with Scikit-HEP's Awkward Array package awkward (Pivarski et al., 2018) through the to_awkward() function, which converts LHE events into columnar data structures optimized for vectorized operations and efficient analysis of large datasets.

89 The library endeavours to support all major Monte Carlo event generators and relies on example
90 XML and HDF5 format files to ensure test coverage.

91 Research impact statement

92 pylhe is regularly used in various research projects and publications within high-energy physics.
93 Notably, it has been cited in Higgs studies (Brehmer et al., 2019; Feuerstake et al., 2025;
94 Stylianou & Weiglein, 2024), in Supersymmetry (SUSY), Beyond the Standard Model (BSM)
95 and dark matter searches (Anisha et al., 2023; Beresford et al., 2024; Beresford & Liu, 2019;
96 Cheung et al., 2024; Kling, 2020; Zhou & Liu, 2022, 2025), and in forward physics studies
97 (Kelly et al., 2022; Kling et al., 2023; Kling & Trojanowski, 2020). It is also employed in
98 methodological studies involving machine learning techniques for event generation and analysis
99 (Brehmer et al., 2020; Kofler et al., 2025).

100 AI usage disclosure

101 Generative AI tools have been used in the development of this software and writing of the
102 manuscript:

- 103 ■ Github's copilot has been used in reviewing pull requests.
- 104 ■ VScode's copilot has been used as an advanced autocomplete.
- 105 ■ ChatGPT has been used to identify the most pythonic solutions in case of ambiguity.

106 All the results generated by these tools have been reviewed by the authors and are correct to
107 the best of our knowledge.

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